

OCPP 1.6-J & 2.1

USER & OPERATOR MANUAL

RELEASE V2.4

OCPP Charge Point Management System

Enterprise EV Charging Network Operations, Ground Plans, Tariffs, Invoicing, RFID & Driver Companion Guide

DOCUMENT TARGET

Charge Point Operators (CPOs), Facility Managers & Drivers

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v2.4 Enterprise Edition

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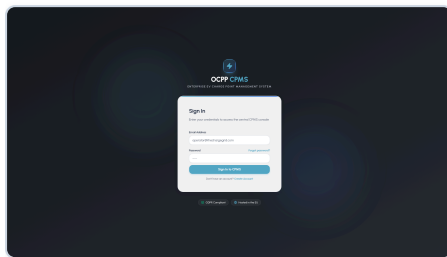
- 16. Mobile Driver Companion Web Application**

1. Getting Started & Authentication Security

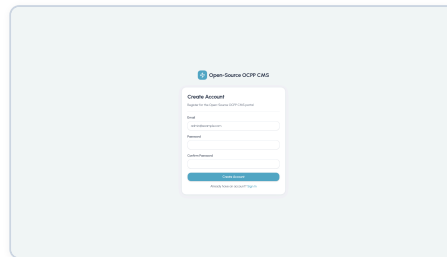
The **OCPP-CPMS** dashboard provides a secure web interface for operating electric vehicle charging infrastructure. Access the platform by navigating to your organization's URL (e.g. <https://ui.mobilitypulse.com> or <http://localhost:3002>).

1.1 Login & Two-Factor Authentication (2FA TOTP)

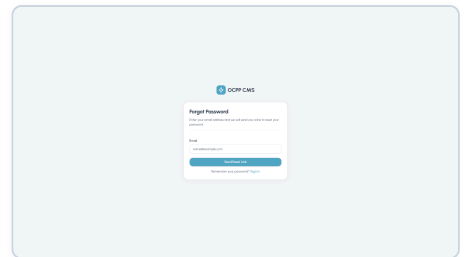
- **Standard Authentication:** Input your registered corporate email and password. Passwords are securely hashed with salted bcrypt.
- **Two-Factor Authentication:** If 2FA is active on your profile, you will be prompted for a 6-digit Time-Based One-Time Password (TOTP) from an authenticator app (Google Authenticator, Microsoft Authenticator, 1Password).
- **Password Reset & Email Verification:** Automated self-service workflows exist for forgotten credentials and initial account email validation.



Secure Operator Login



Driver / User Registration



Self-Service Password Reset

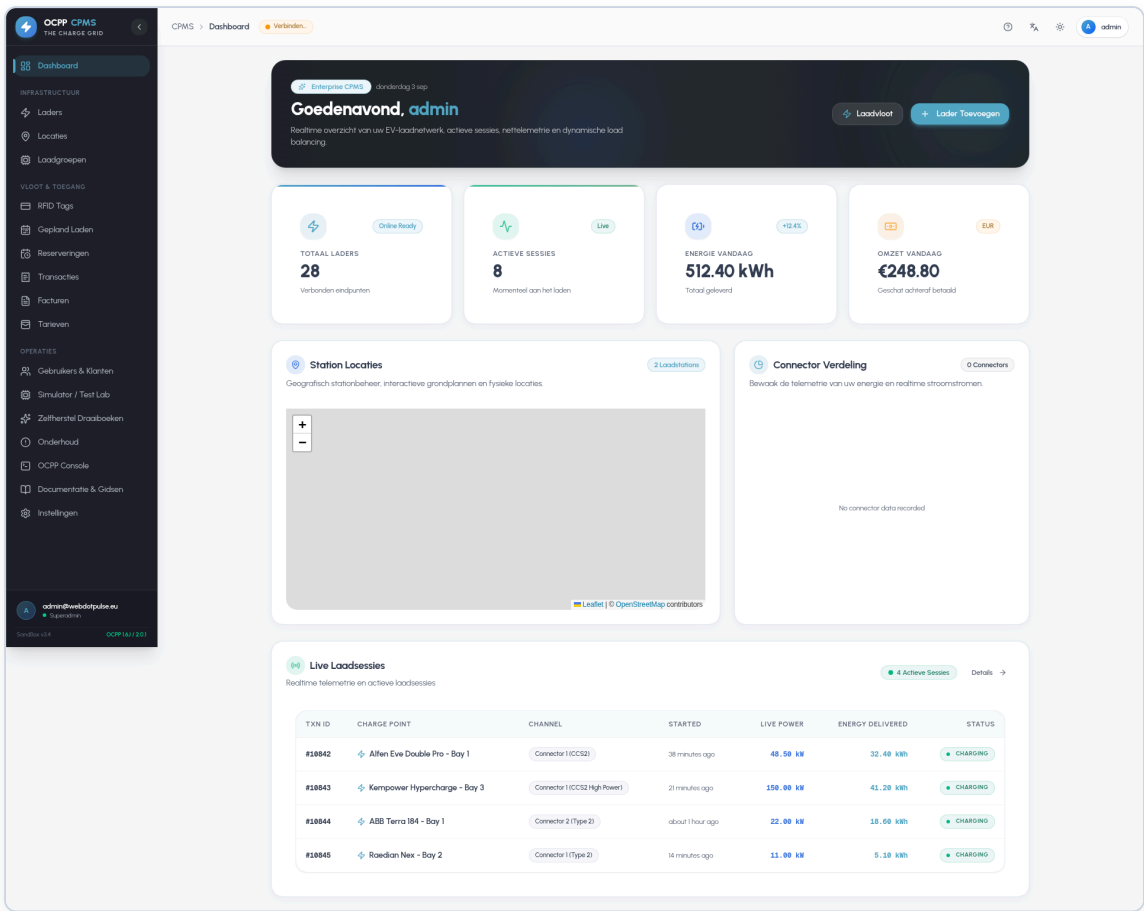
2. Executive Dashboard & Network KPIs

The **Executive Dashboard** (`/dashboard`) provides an instant operational summary across all connected charging stations:

LIVE TELEMETRY OVERVIEW

Metrics update automatically in real-time via Socket.IO WebSocket streams without requiring manual page refreshes.

- **Total Energy Delivered (kWh):** Real-time accumulation of kilowatt-hours dispensed across all active and completed sessions.
- **Active Charging Sessions:** Count of vehicles actively drawing or negotiating power.
- **Fleet Online Health:** Percentage and status breakdown of online vs offline charge points.
- **Geospatial Station Map:** Interactive Leaflet map displaying charging locations with status pins (Available, Charging, Faulted, Offline).
- **24-Hour Load Profiles:** Visual hourly charts detailing peak demand and energy utilization trends.



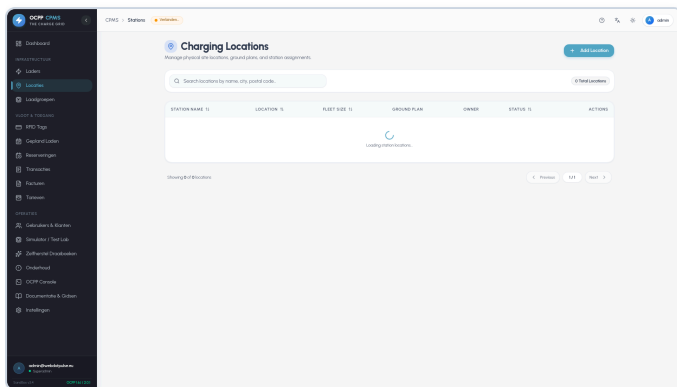
Executive Dashboard with Live Fleet Metrics and Map View

3. Charging Stations & Interactive 2D Ground Plans

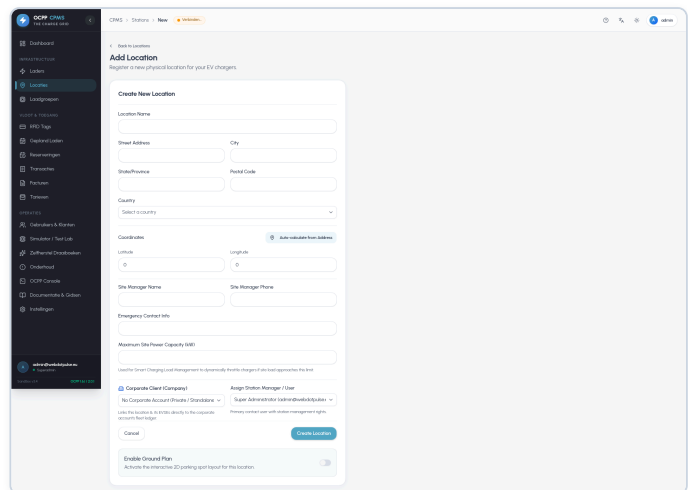
A **Station** represents a physical charging site containing one or more physical charge points with specific geographical, electrical, and operational parameters.

3.1 Station Directory & Management (/stations)

- Create and edit stations with address, GPS latitude/longitude, total power capacity caps, and emergency contacts.
- Filter stations by active status, city, or assigned charge group cluster.



Stations Directory & Geospatial Map

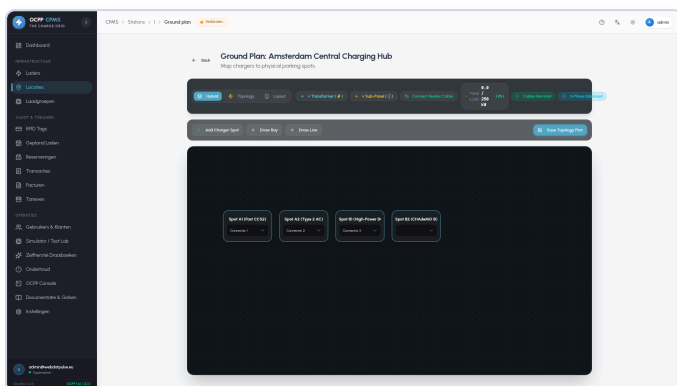


Create Station Modal Form

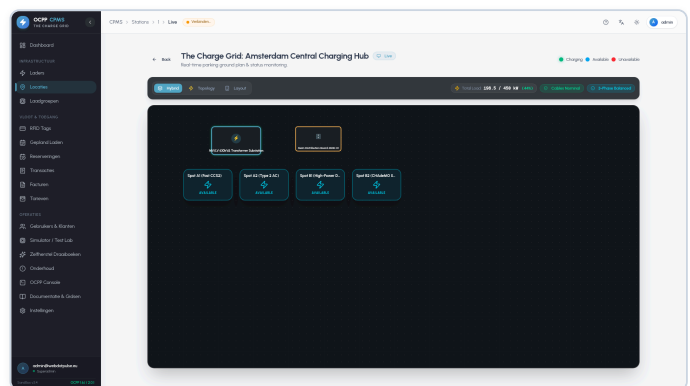
3.2 2D Ground Plan Canvas & Live Floor Monitor

Enable Ground Plan on any station to build high-precision visual parking bay layouts:

- 2D Canvas Editor (/stations/[id]/ground-plan)**: Drag and drop parking bays, rotate bays at 45° angles, draw pedestrian paths and canopy shelters, map physical charger sockets to spots, and draw electrical feeder topology lines.
- Live Floor Monitor (/stations/[id]/live)**: Real-time floor plan showing live spot occupancy, charging wattage, delivered kWh, 3-phase balance (L1/L2/L3), and driver RFID tag badges.



Interactive 2D Ground Plan Canvas Builder

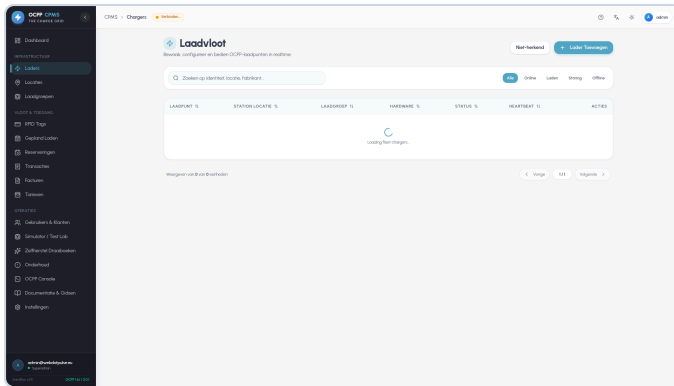


Live Station Floor Plan Monitor with Telemetry

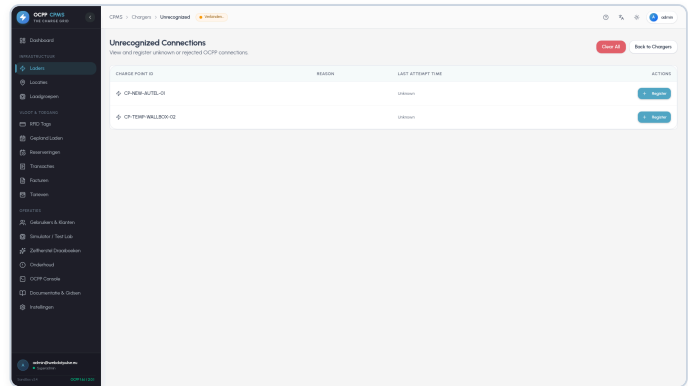
4. Chargers Fleet, EVSE Connectors & Remote Controls

4.1 Fleet Directory (/chargers)

Track all physical charging hardware connected to the central system with OCPP protocol version, firmware build, serial number, and live state.



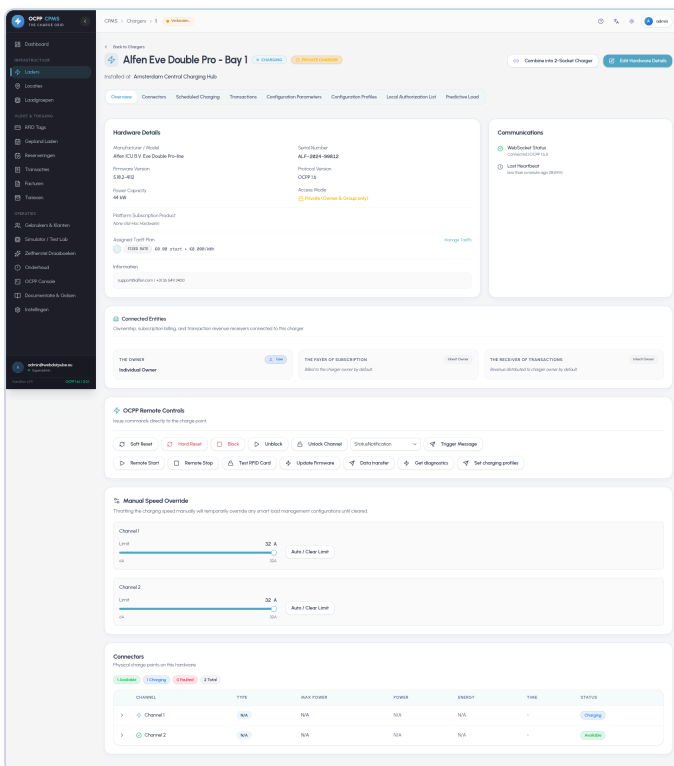
Chargers Fleet Directory



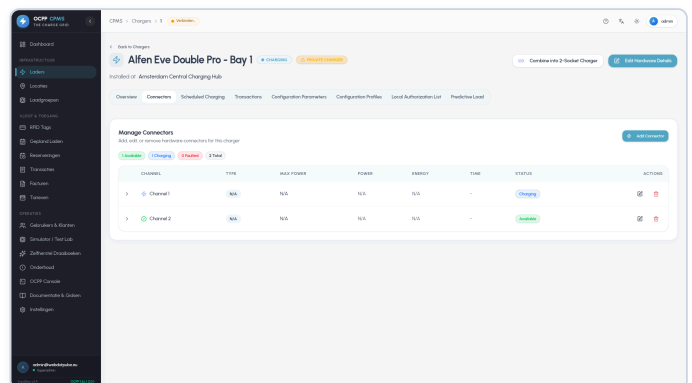
Unrecognized Chargers Auto-Discovery Queue

4.2 Charger Details & Remote Control RPCs (/chargers/[id])

- **Remote Control Commands:** Remote Start Transaction, Remote Stop Transaction, Soft/Hard Reset, Unlock Connector, and Change Availability (Operative/Inoperative).
- **Connectors Tab:** Configure plug types (Type 2, CCS2, CHAdeMO), phase types (1-phase / 3-phase / DC), and max voltage/ampere ratings.
- **Predictive Load Tab:** Inspect 24-hour solar irradiance forecasts and dynamic power limit schedules.



Charger Detail & Remote Control Console



EVSE Connectors Configuration Tab

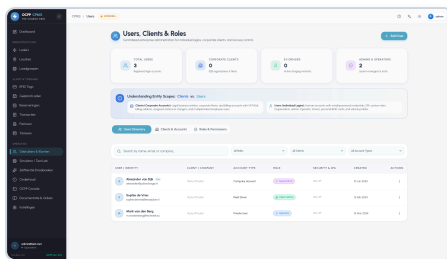
5. User Accounts, Corporate Clients & Granular RBAC

5.1 Corporate B2B Clients vs Individual Users

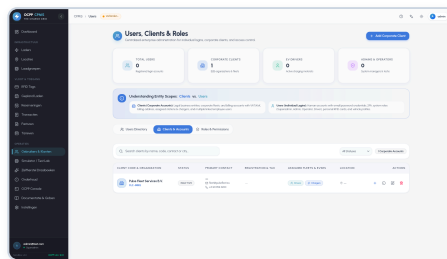
- Corporate Clients:** Business entities holding legal company names, VAT/KvK registrations, billing addresses, SEPA mandates, assigned charging hubs, and employee fleet drivers.
- Individual Users:** Driver accounts with login credentials, 2FA settings, assigned RFID cards, and vehicle battery profiles.

5.2 Roles & Permissions Matrix

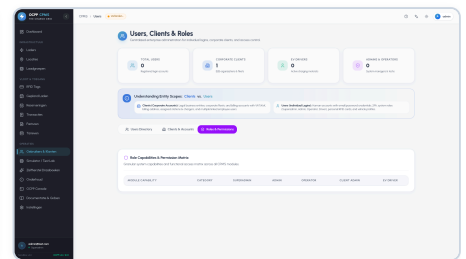
The platform enforces a 5-tier role hierarchy (Superadmin, Platform Admin, Operator/Technician, Client Admin, User/Driver) across all operational modules.



Users Accounts Directory



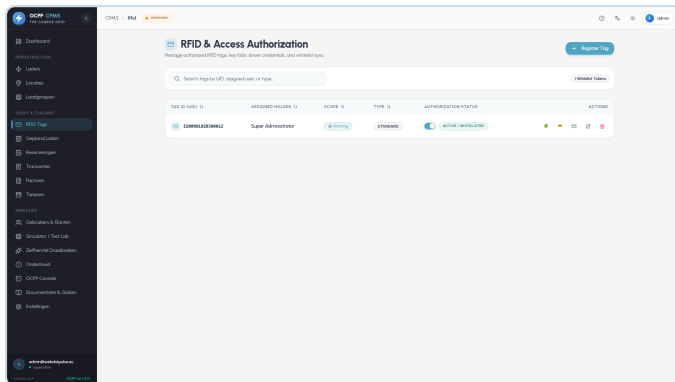
Corporate B2B Clients Hub



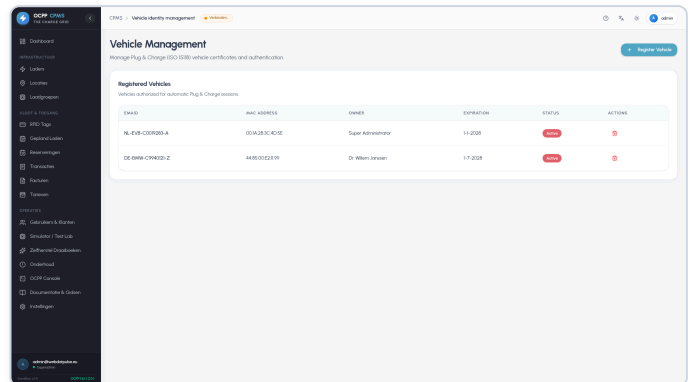
Roles & Capabilities Matrix

5.3 RFID Whitelist & ISO 15118 Plug & Charge

Manage RFID cards by `idTag` with instant hardware cache synchronization, and manage ISO 15118 contract certificates (EMAID) for automated Plug & Charge authorization.



RFID Whitelist Directory



ISO 15118 Plug & Charge Contract Certificates

6.1 Live Active Sessions (/transactions/active)

6.2 Historical Records & Itemized Receipts (/transactions)

[illegible]

Detailed Transaction Tax Receipt

7. Reservations Manager

The **Reservations Manager** (/reservations) allows operators and drivers to schedule exclusive charging windows:

- Reserve specific EVSE sockets for specified arrival times.
- Transmits OCPP **ReserveNow** commands to place physical hardware in **Reserved** state.
- Configurable reservation expiry windows and automatic cancellation on no-show.

OCPP CPMS
THE CHARGE GRID

CPMS > Reservations Verbinden...

Connector Reservations
Dispatch, track, and manage OCPP 1.6 & 2.0.1 / 2.1 hardware connector reservations (ReserveNow / CancelReservation)

Refresh New Reservation

ACTIVE RESERVATIONS 1 CONSUMED (CHARGED) 1 EXPIRED 0 CANCELLED 0

Search by RFID Tag or Charger... Search

Status: All Status

RESERVATION ID	CHARGER & EVSE	RFID / ID TAG	STATUS	EXPIRES AT	CREATED AT	ACTIONS
#501	Allen Eve Double Pro - Bay 1 Connector 1 - Eve Double Pro	E200001928390012 Super Administrator	Active	19:44:11 3-9-2020	3-9-2020 18:59	Cancel
#502	Kempower Hypercharge - Bay 3 Connector 1 - C-Station 400V	CARD-1102-EU	Consumed	18:49:11 3-9-2020	3-9-2020 17:59	

admin@webdatapulse.eu
Super Admin

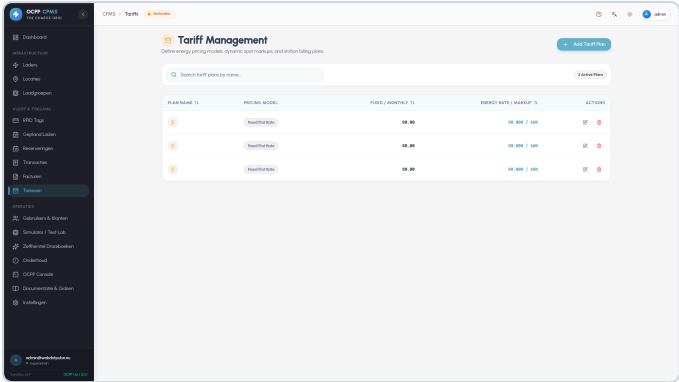
Sandbox v1.4 OCPP 1.6J / 2.0.1

Reservations Management Console

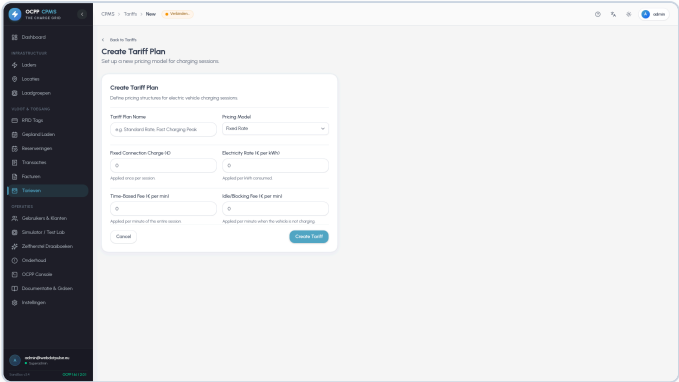
8. Tariffs & Dynamic EPEX Spot Pricing

8.1 4-Part Tariff Pricing Matrices (/tariffs)

FEE COMPONENT	UNIT	DESCRIPTION
Connection Fee	€ / session	Fixed one-time starting charge when session begins.
Energy Fee	€ / kWh	Price per kWh consumed (flat rate or dynamic spot).
Time Fee	€ / hour	Duration charge applied while vehicle is actively plugged in.
Idle Fee	€ / hour	Penalty surcharge after battery reaches 100% SoC.



Tariffs Pricing Structures

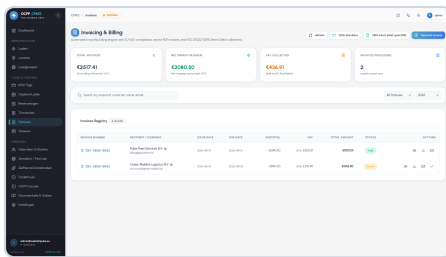


Create Tariff Modal Form

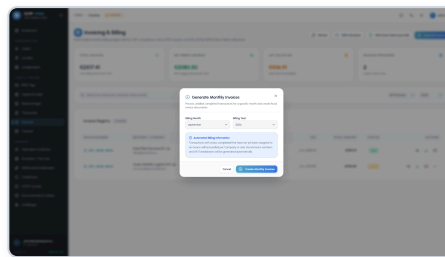
9. Enterprise Invoicing ("Facturen") & ISO 20022 SEPA

The **Invoices Suite** (`/invoices`) provides automated end-of-month billing runs for corporate accounts and private subscribers:

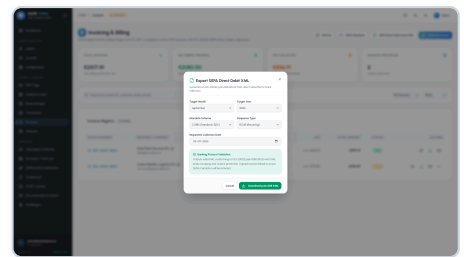
- **Billing Ledger:** Aggregates unbilled charging sessions and displays turnover, subtotal, and 21% VAT metrics.
- **SEPA Direct Debit XML (`pain.008.001.02`):** Generate standardized ISO 20022 banking direct debit files for direct upload to banking portals.
- **SEPA Mandate Management:** Track B2B and CORE Direct Debit mandates with Unique Mandate References (UMR).



Invoicing Ledger & Turnover KPIs



Monthly Batch Invoicing Wizard



ISO 20022 SEPA Direct Debit Export

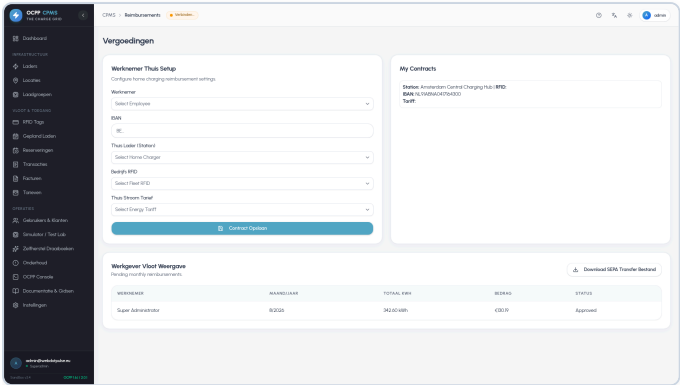
10. Home Reimbursements & Public Payments

10.1 Employee Home Reimbursements (/reimbursements)

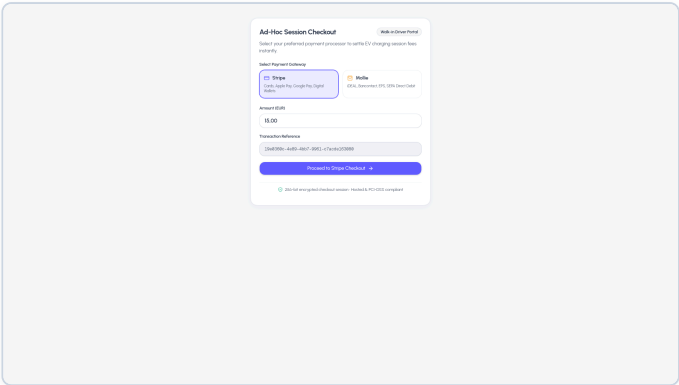
Calculates employee home charging kWh and generates **ISO 20022 SEPA Credit Transfer XML** (`pain.001.001.03`) for bulk corporate reimbursement payouts.

10.2 Public Walk-In Payments (Stripe & Mollie)

Ad-hoc walk-in drivers scan QR codes at charging bays to pay via Stripe (Credit Card, Apple Pay, Google Pay) or Mollie (iDEAL, Bancontact, EPS).



Employee Home Reimbursements & SEPA



Public Ad-Hoc Checkout Portal

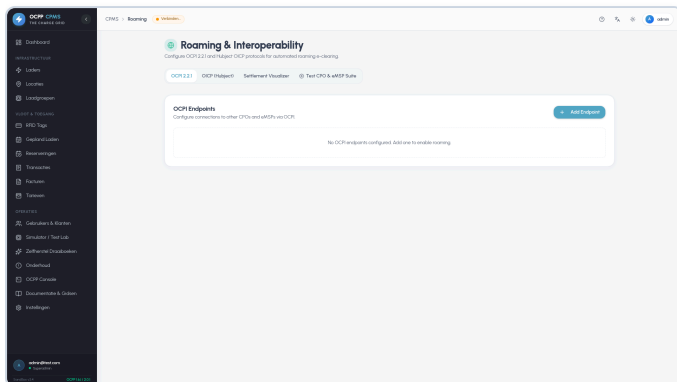
11. Roaming Hubs & V2G Smart Grid

11.1 OCPI 2.2.1 & Hubject OICP Roaming (/roaming)

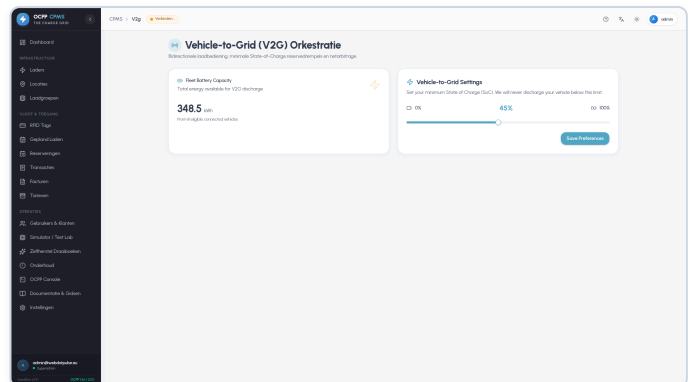
Synchronize locations, tariffs, sessions, tokens, and CDRs with international roaming clearinghouses.

11.2 V2G Battery Orchestration (/v2g)

Manage bidirectional power flow from EV batteries back to local buildings during high-tariff grid hours while maintaining minimum driver SoC reserves.



OCPI 2.2.1 Roaming Hubs

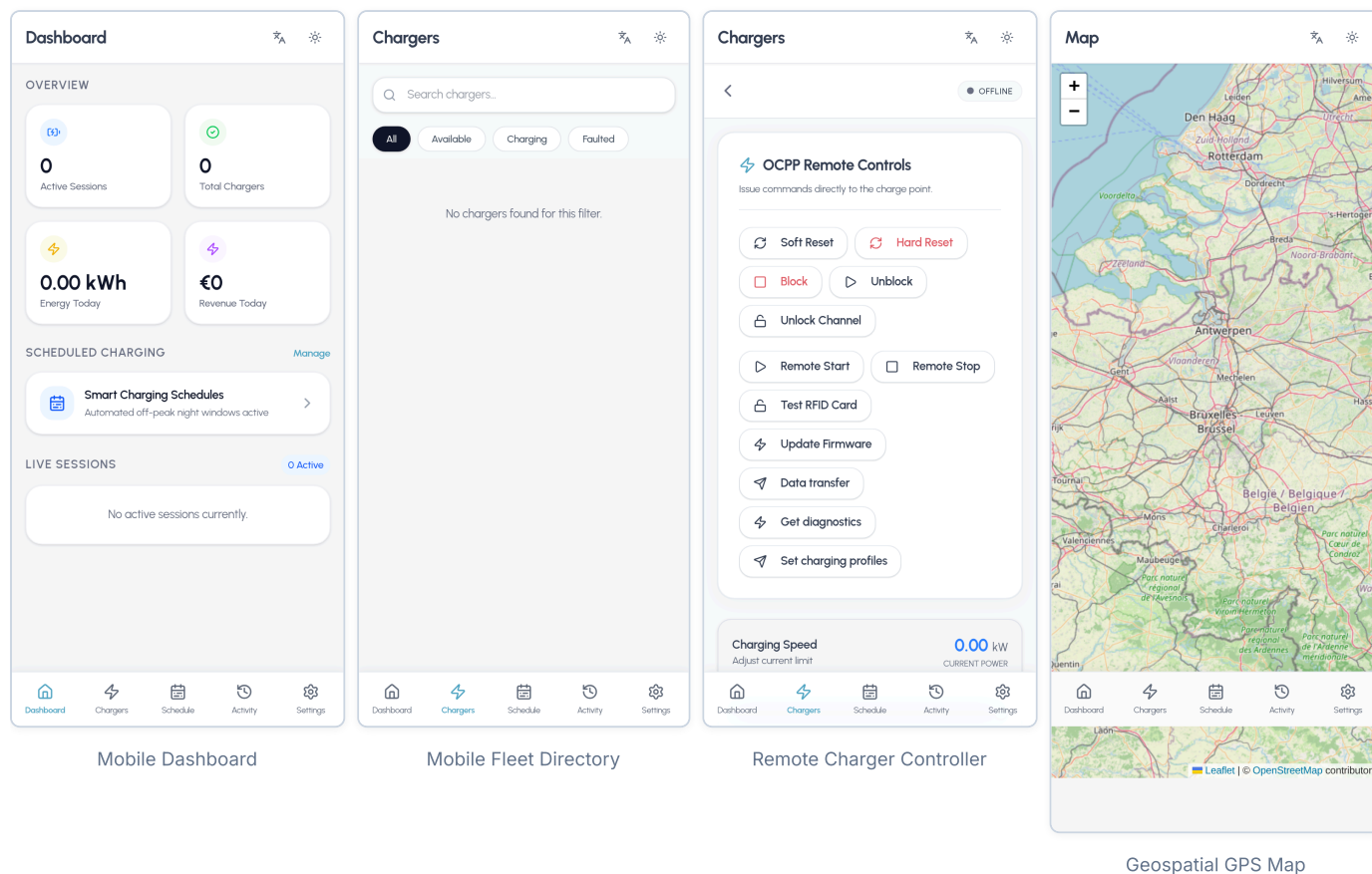


V2G Fleet Battery Orchestration

12. Mobile Driver Companion Web Application

The dedicated mobile web app (/mobile) provides a responsive smartphone experience for drivers on the road:

- **Mobile Dashboard** (/mobile/dashboard): Active charging widget, nearest station finder, and recent receipt history.
- **Mobile Fleet** (/mobile/chargers): List of available chargers with connector plug types and distance.
- **Remote Controller** (/mobile/chargers/[id]): Start and stop charging sessions directly from your phone.
- **Geospatial Map** (/mobile/map): Interactive GPS routing and charger availability pins.



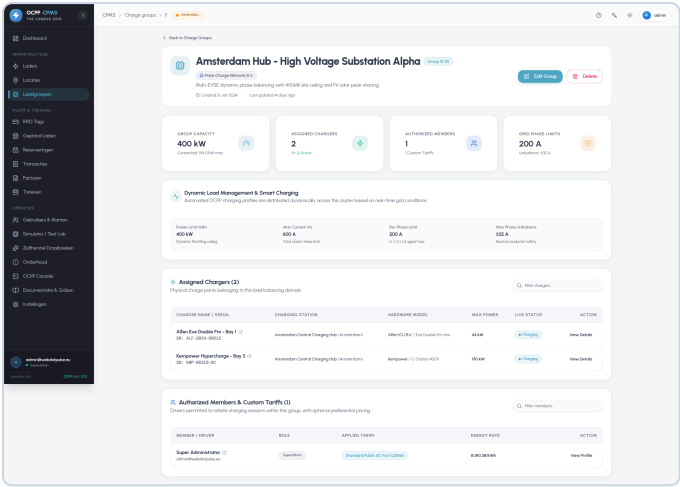
13. Dynamic Load Balancing & Auto-Heal Playbooks

13.1 Dynamic Load Balancing & Phase Allocator (/charge-groups)

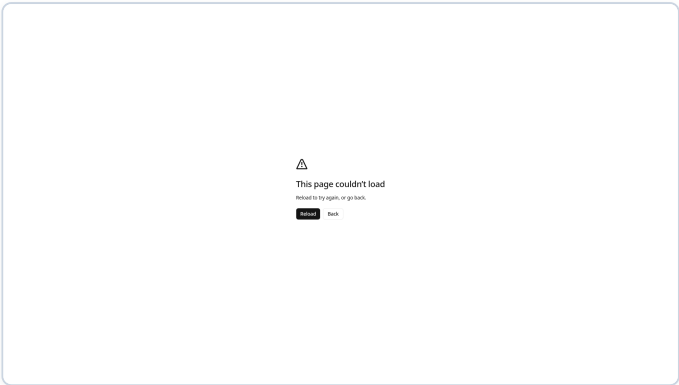
Distribute site electrical capacity, prevent transformer overload, and maintain phase balance across active chargers.

13.2 Automated Recovery Playbooks (/auto-heal-playbooks)

Automate multi-step remediation procedures to resolve connector lock failures, communication timeouts, and mechanical errors.



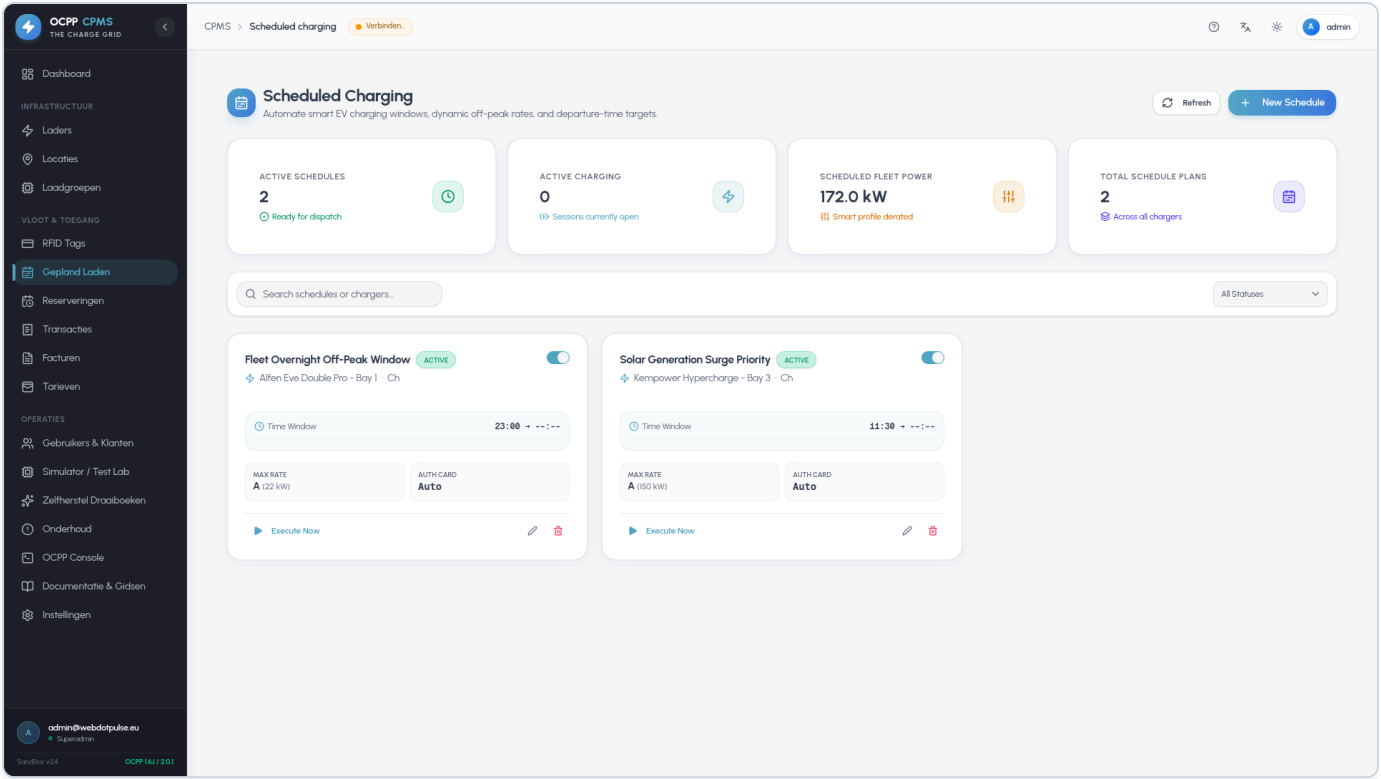
Charge Group Phase Allocation Detail



Automated Recovery Playbooks

13.3 Scheduled Smart Charging Calendar (/scheduled-charging)

Align fleet charging with off-peak spot rates, solar yield windows, and scheduled departure times.



Scheduled Smart Charging Calendar & Time-of-Use Matrix